# Monday LAB Questions (13-oct-2024)

1) Write a program to find the smallest and largest digit in a number using functions.
Implement one function to extract the digits and another function to compare them.
Example:
Input: Enter a number: 4987
Output: Smallest digit: 4, Largest digit: 9

2) Write a recursive program to generate all binary strings of length n. The function should print each binary string in a new line.
Example:
Input: n = 3
Output:
000
001
010
011
100
101
110
111

3) Write a program that checks whether a number is a "strong number". A strong number is a number whose sum of the factorial of digits is equal to the original number. Use multiple functions: one to check if a number is strong, one to calculate the factorial of a digit, and one for input and output.
   Example:
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   Input: Enter a number: 145
   Output: 145 is a strong number (1! + 4! + 5! = 145)

4) Write a recursive function to reverse an array. The function should modify the array in-place.

Example:
Input: Array: [1, 2, 3, 4, 5]
Output: Reversed Array: [5, 4, 3, 2, 1]


5) Given an integer N, generate all possible combinations of N pairs of balanced parentheses using recursion.

Example:
Input: N = 3
Output:  All combinations of 3 pairs of balanced parentheses:
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(()())
(())()
()(())
   ```

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6) Given an array, find the sum of all its subsets using recursion. The function should return the total sum of all possible subsets.

Example:
Input: Array: [3, 1, 2]
Output: Output:  Sum of All Subsets: 24
Explanation: The subsets are: {}, {3}, {1}, {2}, {3, 1}, {3, 2}, {1, 2}, and {3, 1, 2}. The sum is: 0 + 3 + 1 + 2 + 4 + 5 + 3 + 6 = 24.